Unit 5

Multiply Multi-Digit Whole Numbers

Focus Question

How can I multiply multi-digit whole numbers?



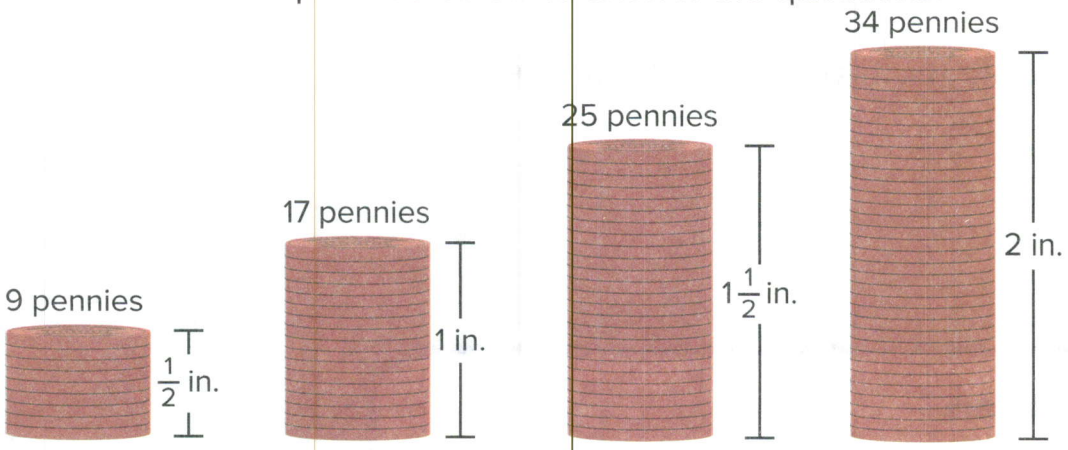
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IGNITE!

Name _____

Mile-High Pennies

Use the stacks of pennies below to answer the questions.



1. How many pennies are in a 1-inch stack? _____

   How many pennies are in a 2-inch stack? _____

2. About how many pennies would be in a 1-foot stack? _____

3. Make a guess: A mile is equal to 5,280 feet. How many pennies do you think would be needed to make a stack that is 1 mile tall?

   _____

   Write a number that is definitely too large for a reasonable estimate. _____

   Write a number that is definitely too small for a reasonable estimate. _____

4. What is your calculated estimate for the number of pennies in a stack that is 1 mile tall? _____

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Lesson 5-1

Understand Powers and Exponents

Be Curious

Which Doesn't Belong?

7×7

4×4

6×8

10×10

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Math is... Mindset

What do you want your classmates to know about your math story?

Learn

At Week 1, Dean had 10 pennies Each week after, Dean increased the number of pennies by 10 times the previous week.

During which week will Dean have 1,000,000 pennies?

You can organize the information in a table to help determine the solution.

Math is... Patterns

What patterns do you notice in the table?

Week	Multiplication Expression	Number of Pennies Added each Week
1	10	10
2	10×10	100
3	$10 \times 10 \times 10$	1,000
4	$10 \times 10 \times 10 \times 10$	10,000
5	$10 \times 10 \times 10 \times 10 \times 10$	100,000
6	$10 \times 10 \times 10 \times 10 \times 10 \times 10$	1,000,000

A **power of 10** is the product of 10 multiplied by itself a number of times.

Dean will have 1,000,000 pennies in Week 6.

You can write a power of 10 as a multplication expression with factors of 10.

You can also write a power of 10 in **exponential form** using a **base** and an **exponent**.

$\underbrace{10 \times 10}_{\text{2 factors}} = 10^{2}$ ← exponent

↑ base

Work Together

Write 10^{8} as a multiplication expression. Then, find the product.

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On My Own

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Name _____

Write the exponential form as a multiplication expression.

1. 10^4

2. 10^2

3. 10^3

4. 10^6

Write the exponential form.

5. $10 \times 10 \times 10 =$ _____

6. $10 \times 10 \times 10 \times 10 \times 10 =$ _____

7. $10 \times 10 \times 10 \times 10 =$ _____

8. $10 \times 10 =$ _____

Write the exponential form of each power of 10.

9. $10 =$ _____

10. $1{,}000 =$ _____

11. $100 =$ _____

12. $10{,}000 =$ _____

13. Rachel finds the value of 10^5 as shown. Do you agree with her solution? Tell why.

$10^5 = 10 \times 5 = 50$

14. **STEM Connection** Grace reviewed 10^6 lines of a computer program. How many lines did she review? Write the product.



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15. Trevor's personal walking goal is shown on his activity tracker.



A. How can you help Trevor write this goal using an exponent?

B. How can you help Trevor write this goal as a product of 10s?

16. Jenny's father gives her $10. Her grandfather offers to give her ten times the value her father gave her. Her grandmother offers her a choice of either ten times the value her grandfather offered or $500. Which of her grandmother's offers should Jenny choose? Explain.

17. Extend Your Thinking Consider the inequality shown.

$$10 \times 10 \times 10 \times 10 < b < 10^6$$

What is the value of b? Explain.

Reflect

What patterns did you notice when writing different forms of powers of 10?

Math is... Mindset

How have you told your classmates about your math story?

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Be Curious

How are they the same?
How are they different?

$15 \times 10{,}000$

$15 \times 10 \times 10 \times 10 \times 10$

15×10^4

150,000

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Math is... Mindset

What do you do to avoid getting distracted?

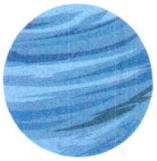
Learn

The distances from these planets to the Sun are shown as multiplication expressions.

How can you determine the value of these expressions?



Mercury
about 36×10^6 mi



Neptune
about 3×10^9 mi

First, determine the distance from Mercury to the Sun. Look for pattterns when multiplying by a power of 10.

$36 \times 10^6 = 36 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10$

$= 36 \times 1{,}000{,}000$

$= 36{,}000{,}000$

The exponent is the same as the number of zeros in the product.

The distance from Mercury to the Sun is about **36,000,000 miles.**

You can use patterns to determine the distance from Neptune to the Sun.

$3 \times 10^9 = 3 \times 1{,}000{,}000{,}000$

$= 3{,}000{,}000{,}000$

The distance from Neptune to the Sun is about 3,000,000,000 miles.

Math is... Structure

Why does the place of the digits in a number shift each time you multiply by 10?

When multiplying by powers of 10, there is a pattern in the number of zeros in the product in relationship to the exponent.

Work Together

Find the value of each expression. Explain how you used patterns to help you.

32×10^2 32×10^3 32×10^4

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On My Own

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Name _____

What is the product? Use patterns to solve.

1. $12 \times 10 =$ _____
   $12 \times 100 =$ _____
   $12 \times 1{,}000 =$ _____

2. $24 \times 1{,}000 =$ _____
   $24 \times 10{,}000 =$ _____
   $24 \times 100{,}000 =$ _____

3. $33 \times 10^2 =$ _____
   $33 \times 10^3 =$ _____
   $33 \times 10^4 =$ _____

4. $57 \times 10^4 =$ _____
   $57 \times 10^5 =$ _____
   $57 \times 10^6 =$ _____

What is the product?

5. 23×10^3

6. 581×10^2

7. 60×10^4

8. 103×10^2

What is the unknown factor?

9. $571 \times$ _____ $= 5{,}710$

10. $43 \times$ _____ $= 4{,}300{,}000$

11. $6 \times$ _____ $= 6{,}000$

12. $28 \times$ _____ $= 280{,}000$

13. How can you describe the relationship between the equations shown?

$6 \times 10^5 = 600{,}000$
$6 \times 10^7 = 60{,}000{,}000$
$6 \times 10^9 = 6{,}000{,}000{,}000$

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14. Error Analysis Carol says the equation that she wrote is correct. How do you respond to her?

$80 \times 10{,}000 = 80{,}000$

15. Which equations are *true*? Circle all that apply.

A. $6 \times 100 = 6 \times 10 \times 10 \times 10$

B. $10{,}000 \times 4 = 10 \times 10 \times 10 \times 10 \times 4$

C. $15 \times 10^3 = 1{,}500$

D. $70 \times 10 \times 10 = 7{,}000$

16. Extend Your Thinking Find the unknown factor that is a whole number. Explain your thinking.

$$? \times 10^5 = 56{,}300{,}000$$

Reflect

What patterns did you notice when multiplying by powers of 10?

Math is... Mindset

How have you avoided getting distracted?

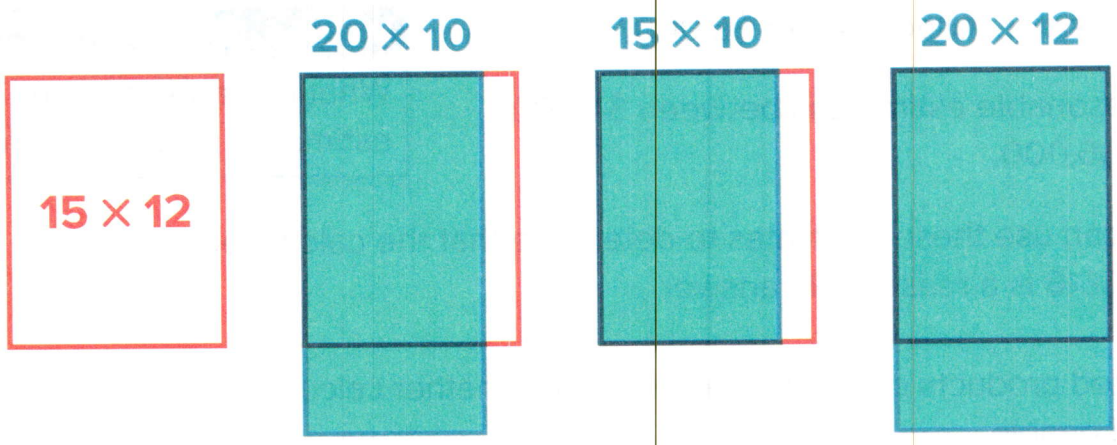
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Lesson 5-3

Estimate Products of Multi-Digit Factors

Be Curious

What do you notice?
What do you wonder?



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Math is... Mindset

What helps you make sense of a situation?

Learn

On Saturday, 432 people go to the theater.

About how much money does the theater collect on Saturday?



You can use strategies you know to determine a reasonable estimate.

One Way Compatible numbers

432×13

$400 \times 15 = 4 \times 100 \times 15$
$= 4 \times 15 \times 100$
$= 6{,}000$

The theater collects about $6,000.

Another Way Rounded factors

432×13

$430 \times 10 = 4{,}300$

The theater collects about $4,300.

A reasonable estimate is between $4,300 and $6,000.

You can use these estimates to determine that the calculated solution of $5,616 is a reasonable answer.

Math is... Choosing Tools

What can and can't an estimated product tell you?

Estimated products can help you determine whether calculations are reasonable.

Work Together

Estimate the product of 879×36.

Which strategy did you use? Explain why.

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On My Own

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Name _____

Estimate the product.

1. 643 × 18

2. 325 × 62

3. 438 × 27

4. 572 × 49

5. On a school trip, 54 students went to a museum. Each ticket cost $23. About how much did all students spend on tickets?

6. The town library has 478 shelves. Each shelf holds 38 books. About how many books does the library have?

7. A vendor at a fair is selling her paintings for $23 each. Over the course of the fair, 339 people purchase her paintings. About how much did the vendor earn over the course of the fair?

8. The fifth graders sold 405 baked goods at the bake sale. About how much did the fifth graders earn?



9. **Error Analysis** Han estimates that the product of 492 and 32 will be 1,200. How do you respond to Han?

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10. Which equation represents a reasonable estimate for 658 × 19? Explain.

    A. 700 × 10 = 7,000

    B. 650 × 20 = 13,000

    C. 600 × 10 = 6,000

11. If you estimate the product of 246 × 38, will the estimate be greater using rounded numbers or compatible numbers? Why?

12. **Extend Your Thinking** A recycling club has a goal of collecting 8,000 plastic bottles. Each of the 26 students in the club collected 72 bottles a day for 5 days. About how many bottles did they collect at the end of 5 days? Did they meet their goal?

Reflect

How can you use estimates to determine whether a calculated product is reasonable?

Math is... Mindset

How have you made sense of a situation?

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Lesson 5-4

Use Area Models to Multiply Multi-Digit Factors

Be Curious

What do you see?



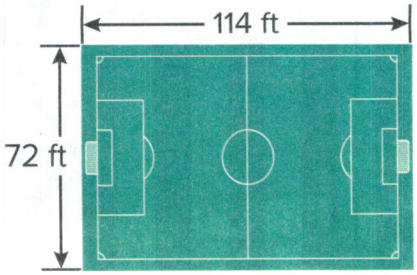
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Math is... Mindset

How can you show that you understand your partner's point of view?

Learn

How can you determine the area of the youth soccer field?



You can use an area model to solve $72 \times 114 = A$.

Decompose the factors by place value.

	100	+ 10	+ 4
70			
+ 2			

Determine partial products.

	100	+ 10	+ 4
70	7,000	700	280
+ 2	200	20	8

Math is... Modeling

How does an area model help you understand multiplication?

Add the partial products to determine the product.

$7{,}000 + 700 + 280 + 200 + 20 + 8 = 8{,}208$

The area of the soccer field is 8,208 square feet.

You can use area models to multiply multi-digit factors.

Work Together

Use an area model and partial products to determine the product of 304×68.

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On My Own

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Name _____

Complete the area model. Then solve to find the product.

1.

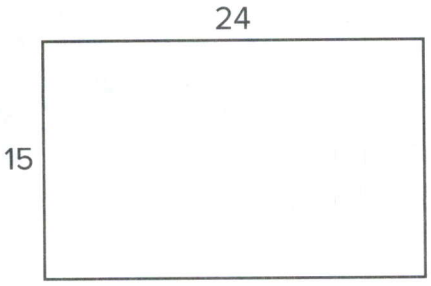
	100	+ 50	+ 4
30	3,000		
+ 6	600		

2.

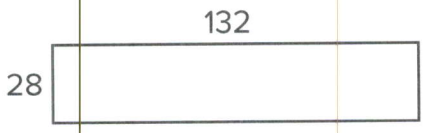
	400	+ 20	+ 1
50	20,000		50
+ 2			

What is the product? Use area models to solve.

3. $15 \times 24 =$ _____



4. $28 \times 132 =$ _____



5. $33 \times 78 =$ _____

6. $72 \times 225 =$ _____

Write the multiplication equation based on the area model. Then solve to find the product.

7.

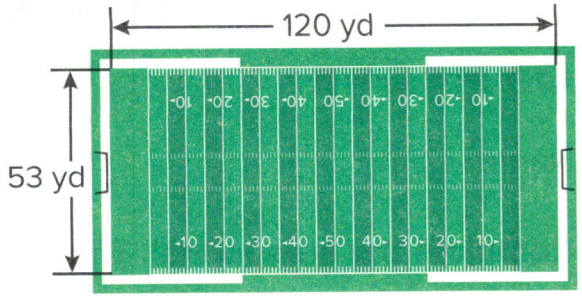
	200	+ 10	+ 8
20			
+ 8			

8.

	400	+ 10	+ 6
60			
+ 3			

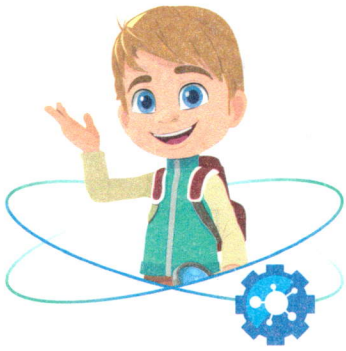
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9. What is the area of the football field?



10. A school collected 128 boxes of canned food. Each box has 45 cans. How many cans of food did the school collect?

11. **STEM Connection** Owen has 14 sketchbooks that each contain 128 sketches of insects. How many sketches of insects does Owen have in all?



12. **Extend Your Thinking** How could you use an area model to determine the product of 452 × 273? Explain.

Reflect

How did I think like a mathematician to multiply multi-digit factors?

Math is... Mindset

How have you shown that you understand your partner's point of view?

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Lesson 5-5

Use Partial Products to Multiply Multi-Digit Factors

Be Curious

What do you notice?
What do you wonder?

	300	+ 70	+ 4
40	12,000	2,800	160
+ 3	900	210	12

$$
\begin{array}{rl}
374 & \\
\times\ 43 & \\
\hline
12{,}000 & \leftarrow 40 \times 300 \\
2{,}800 & \leftarrow 40 \times 70 \\
160 & \leftarrow 40 \times 4 \\
900 & \leftarrow 3 \times 300 \\
210 & \leftarrow 3 \times 70 \\
+\ 12 & \leftarrow 3 \times 4 \\
\hline
16{,}082 &
\end{array}
$$

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Math is... Mindset

What steps can you take to focus on your work today?

Learn

A stadium has 164 rows of seats.

How many seats are in the stadium?



You can use partial products to solve the problem.

Decompose the factors by place value.

$164 = 100 + 60 + 4$

$82 = 80 + 2$

Determine the partial products.

$$\begin{array}{rrl} & 164 & \\ \times & 82 & \\ \hline & 8{,}000 & \leftarrow 80 \times 100 \\ & 4{,}800 & \leftarrow 80 \times 60 \\ & 320 & \leftarrow 80 \times 4 \\ & 200 & \leftarrow 2 \times 100 \\ & 120 & \leftarrow 2 \times 60 \\ + & 8 & \leftarrow 2 \times 4 \\ \hline & 13{,}448 & \end{array}$$

Multiply 80 by 100, 60, and 4.

Multiply 2 by 100, 60, and 4.

Add the partial products.

There are 13,448 seats in the stadium.

Math is... Generalizations

How does the area model help you understand how this strategy works?

Partial products is another multiplication strategy you can use to multiply multi-digit whole numbers.

Work Together

What is the product? Use partial products to solve. Show 142 × 63 vertically.

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On My Own

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Name _____

Find the unknown partial products. Then find the product.

1.

$$\begin{array}{r} 325 \\ \times \quad 73 \\ \hline 21{,}000 \\ 1{,}400 \\ 350 \\ \square \\ \square \\ + \quad \square \\ \hline \square \end{array}$$

2.

$$\begin{array}{r} 104 \\ \times \quad 28 \\ \hline 32 \\ 800 \\ \square \\ + \quad \square \\ \hline \square \end{array}$$

What is the product? Use partial products to solve.

3. 17×86

4. 24×129

5. 36×93

6. 222×58

7. A sporting goods store sold 24 mountain bikes. How much money did they make selling bikes?



8. The store also sold 12 mountain bike and scooter packages each for $367. How much money did they make?

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9. At a school fundraiser, 327 donors give $25 each. How much money does the school collect?

10. A store has 60 boxes of shirts with 152 shirts in each box. How many shirts are there in all?

11. **Error Analysis** Raya used partial products to find the product of 128 × 17. What can you say about Raya's work?

```
  208
×  17
-----
2,000
  100
   80
1,400
   70
+  56
-----
3,706
```

12. **Extend Your Thinking** Solve using partial products. Explain your answer.

384 × 725

Reflect

How can you use partial products to find the product of multi-digit factors?

Math is... Mindset

What steps did you take to focus on your work today?

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Lesson 5-6

Relate Partial Products to an Algorithm

Be Curious

What do you notice?
What do you wonder?

$$\begin{array}{r} 132 \\ \times \quad 3 \\ \hline 300 \leftarrow 3 \times 100 \\ 90 \leftarrow 3 \times 30 \\ + \quad 6 \leftarrow 3 \times 2 \\ \hline 396 \end{array}$$

$$\begin{array}{r} 132 \\ \times \quad 3 \\ \hline 396 \end{array}$$

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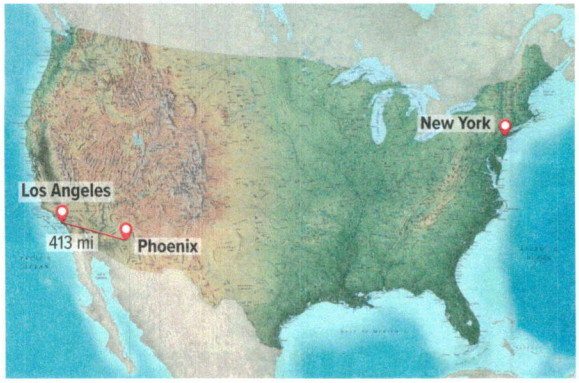
Math is... Mindset

How can you be part of the classroom community?

Learn

The distance from Los Angeles to New York City is 7 times as far as from Los Angeles to Phoenix.

How can you determine the distance from Los Angeles to New York City?



You can multiply using an **algorithm**.

Step 1 Multiply the ones.

$7 \times 3 = 21$

Regroup 21 as 2 tens and 1 one.

Show 2 tens.

Show 1 one.

$$\begin{array}{r} {\scriptstyle 2} \\ 413 \\ \times \quad 7 \\ \hline {\scriptstyle 1} \end{array}$$

Step 2 Multiply the tens.

$7 \times 10 = 70$

Add the 2 tens from Step 1.

$70 + 20 = 90$

Show 9 tens.

$$\begin{array}{r} {\scriptstyle 2} \\ 413 \\ \times \quad 7 \\ \hline 91 \end{array}$$

Step 3 Multiply the hundreds.

$7 \times 400 = 2{,}800$

Show 28 hundreds.

The distance from Los Angeles to New York City is 2,891 miles.

$$\begin{array}{r} {\scriptstyle 2} \\ 413 \\ \times \quad 7 \\ \hline 2{,}891 \end{array}$$

Math is... Generalizations

How are the partial products strategy and this algorithm related?

Work Together

Find the product using an algorithm.

$$\begin{array}{r} 3{,}021 \\ \times \quad 4 \\ \hline \end{array}$$

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On My Own

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Name _____

What is the product?

1. $\begin{array}{r} 327 \\ \times \quad 6 \\ \hline \end{array}$

2. $\begin{array}{r} 543 \\ \times \quad 8 \\ \hline \end{array}$

3. $\begin{array}{r} 1{,}931 \\ \times \quad 5 \\ \hline \end{array}$

4. $\begin{array}{r} 3{,}462 \\ \times \quad 4 \\ \hline \end{array}$

What is the product? Choose the correct answer.

5. $188 \times 7 = s$

   A. $s = 1{,}300$

   B. $s = 1{,}316$

   C. $s = 1{,}388$

   D. $s = 1{,}406$

6. $237 \times 9 = v$

   A. $v = 2{,}033$

   B. $v = 2{,}163$

   C. $v = 2{,}153$

   D. $v = 2{,}133$

7. $2{,}623 \times 2 = y$

   A. $y = 5{,}246$

   B. $y = 4{,}246$

   C. $y = 5{,}126$

   D. $y = 5{,}616$

8. $5{,}246 \times 3 = r$

   A. $r = 15{,}882$

   B. $r = 16{,}838$

   C. $r = 16{,}612$

   D. $r = 15{,}738$

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9. **STEM Connection** Hiro knows that Lake Michigan is 922 feet deep. He knows that the Atlantic Ocean is 28 times as deep. How deep is the Atlantic Ocean?



10. Joey earns $355 a week mowing lawns. How much money does he earn in 6 weeks?

11. The pedometer shows the number of steps Maria walks each day. How many steps did she walk after 7 days?



12. On a road trip, Emily and her family listen to 3 podcasts. Each one is 45 minutes long. For how many seconds do they listen to podcasts?

13. **Extend Your Thinking** Explain why it is important to follow each step of the algorithm when multiplying.

Reflect

How are partial products and an algorithm for multiplication related?

Math is... Mindset

How have you been part of the classroom community?

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Unit 5

Multiplication of 2-Digit Numbers

CHERYL TOBEY MATH PROBES

Name _____

Four students were in the process of finding this product:

42×13.

Decide if each student's approach is a correct way to find the product. Circle Yes or No.

Do not actually complete the process to find the product.

Student 1

	40	+ 2
10	400	20
+ 3	120	6

Circle Yes or No.

Yes No

Explain why you chose Yes or No.

Student 2

$(40 \times 20) + (2 \times 3)$

Circle Yes or No.

Yes No

Explain why you chose Yes or No.

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Student 3

$13 \times 40 + 13 \times 2$

Circle Yes or No.

Yes No

Explain why you chose Yes or No.

Reflect On Your Learning

I'm confused. I'm still learning. I understand. I can teach someone else.



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Lesson 5-7

Multiply Multi-Digit Factors Fluently

Be Curious

How are they the same?
How are they different?

443 × 15 2,000 200 15 4,000 400 + 30 6,645	21 443 × 15 2,215 + 4,430 6,645

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Math is... Mindset

What makes you feel excited when doing math?

Learn

Last weekend, a store sold 549 T-shirts.

How can you determine how much money a store made from selling T-shirts last weekend?



You can use an algorithm to solve the problem.

Step 1 Multiply 549 × 6.

$$\begin{array}{r} {\scriptstyle 2\ 5} \\ 549 \\ \times\ 26 \\ \hline 3{,}294 \end{array}$$

Math is... Generalizations

How might using this algorithm be different when multiplying two 3-digit numbers?

Step 2 Multiply 549 × 20.

$$\begin{array}{r} {\scriptstyle 1} \\ 549 \\ \times\ 26 \\ \hline 3{,}294 \\ 10{,}980 \end{array}$$

Step 3 Add partial products.

$$\begin{array}{r} 549 \\ \times\ 26 \\ \hline 3{,}294 \\ +10{,}980 \\ \hline 14{,}274 \end{array}$$

The store made $14,274 from selling T-shirts last weekend.

An algorithm can be a more efficient way to multiply.

Work Together

What is the product?

$$\begin{array}{r} 2{,}165 \\ \times\ 34 \\ \hline \end{array}$$

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On My Own

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Name _____

What is the product?

1. 327 × 15

2. 2,682 × 35

3. 135 × 23

4. 792 × 11

5. 180 × 72

6. 378 × 29

7. 623 × 64

8. 1,432 × 73

9. The students at Mia's school use about 3,408 sheets of paper per day. How many sheets of paper do the students use after 25 days of school?

10. In one video game, you can build a house for every 375 points you earn. Jack built 23 houses. How many points did he earn?

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11. On a road trip, a family traveled 412 miles each day. How many miles had they traveled by the end of 5 days?

12. A hotel room costs $193 per night. How much will a 7-day stay cost?

13. **STEM Connection** Adult dogs have 42 teeth. Over the course of a week, Ruby checks the teeth of 65 dogs. How many teeth does she check in all?



14. **Extend Your Thinking** How does knowing how to solve multiplication equations using partial products help you solve multiplication equations using an algorithm?

Reflect

Why might using an algorithm be more efficient than using partial products when multiplying?

Math is... Mindset

What has made you feel excited when doing math?

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Unit Review Name _____

Vocabulary Review

Choose the correct word(s) to complete each sentence.

algorithm	exponential form
base	partial products
exponent	power of 10
estimate	round

1. When multiplying multi-digit numbers, you can decompose the factors and find _____. Then, you add the _____ to get the product. (Lesson 5-4)

2. The _____ of a number written in exponential form tells you how many times the base is multiplied by itself. (Lesson 5-1)

3. The _____ of a number includes a base raised to a power called the _____. (Lesson 5-1)

4. To estimate a solution, you can _____ the values used before performing operations. (Lesson 5-3)

5. A(n) _____ is written in exponential form using 10 as the base and the number of times 10 is multiplied by itself as the power, or exponent. (Lesson 5-1)

6. A(n) _____ is an approximate solution to a problem. (Lesson 5-3)

7. A(n) _____ is a set of steps used to solve a problem. (Lesson 5-6)

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Review

8. Which expression or value is equivalent to 10^4? (Lesson 5-1)

A. 1,000

B. 10×4

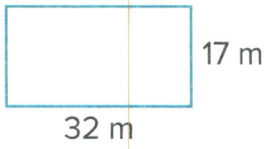
C. $10 \times 10 \times 10 \times 10$

D. $10 + 10 + 10 + 10$

9. The rock museum has 324 display drawers. Each drawer holds 23 rock samples. About how many rock samples does the museum have? (Lesson 5-3)

There are about _____ rock samples.

10. Use an area model to find the area of the outdoor recreational center. (Lesson 5-4)



The area is _____ square meters.

11. Complete the partial products to find 328×14. (Lesson 5-5)

$$\begin{array}{r} 328 \\ \times\ 14 \\ \hline 3{,}000 \leftarrow 10 \times 300 \\ \square \leftarrow 10 \times 20 \\ \square \leftarrow 10 \times 8 \\ 1{,}200 \leftarrow 4 \times 300 \\ 80 \leftarrow 4 \times 20 \\ \square \leftarrow 4 \times 8 \end{array}$$

$328 \times 14 =$ _____

12. Blaire reads 242 pages each week. How can she use partial products or an algorithm to find the number of pages she reads in 6 weeks? (Lesson 5-6)

_____

_____

_____

_____

13. Find the product using an algorithm. (Lesson 5-7)

$429 \times 31 =$ _____

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14. Which equation or equations are true? Select all. (Lesson 5-2)

A. $7 \times 100 = 7 \times 10 \times 10 \times 10$

B. $2 \times 1{,}000 = 2 \times 10^3$

C. $50 \times 10 \times 10 \times 10 = 50{,}000$

D. $8 \times 10^3 = 8 \times 10 \times 10$

E. $16 \times 10^2 = 160$

15. Which equation represents the best estimate for 367×29? (Lesson 5-3)

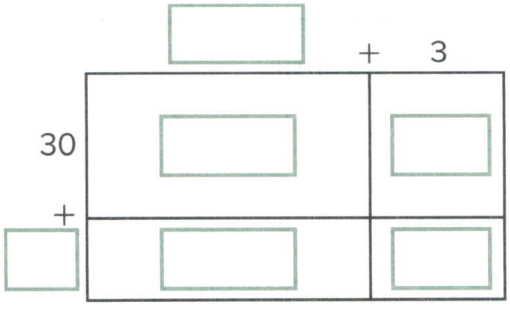
A. $300 \times 20 = 6{,}000$

B. $300 \times 30 = 9{,}000$

C. $400 \times 20 = 8{,}000$

D. $400 \times 30 = 12{,}000$

16. Fill in the area model and use partial products to find 53×37. (Lesson 5-5)



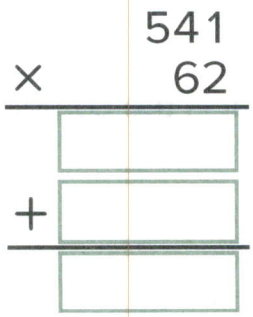
$53 \times 37 =$ _____

17. The track team runs a combined average of 126 miles per day. How many miles might the team run in 5 weeks? Estimate the solution. Then solve. (Lesson 5-7)

estimation: _____ miles

solution: _____ miles

18. Complete the algorithm to find the product. (Lesson 5-7)



19. Write 10^6 as a product of 10s. Then write the product. (Lesson 5-2)

$10^6 =$ _____

$=$ _____

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Performance Task

Owen is studying how much honey is produced by honey bees at two different locations. The hives at location A house 13×10^2 honey bees and produce about 121 ounces of honey each week. The hives at location B house 27×10^2 honey bees and produce about 344 ounces of honey each week.

Part A: What is the difference between the number of honey bees at each location?

Part B: What is the combined amount of honey produced each year?

Reflect

What are some different strategies you can use to multiply multi-digit numbers?

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Unit 5

Fluency Practice

Name _____

Fluency Strategy

You can choose a strategy to add. You can adjust the numbers, use partial sums, or use an algorithm.

Adjust the numbers.

$$1{,}521 + 2{,}299$$

$$\boxed{-1} \qquad \boxed{+1}$$

$$1{,}520 + 2{,}300 = 3{,}820$$

1. Adjust the numbers to find the sum.

$$2{,}413 + 1{,}097$$

$$\boxed{-} \qquad \boxed{+}$$

_____ + _____ = _____

Fluency Flash

What is the sum?

2.

	thousands	hundreds	tens	ones
	1	0	9	7
+	2	4	5	6

3.

	thousands	hundreds	tens	ones
	4	2	7	9
+	2	3	6	8

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Fluency Check

What is the sum or difference?

4. 3,241 + 316 = _____

5. 987 – 453 = _____

6. 824 – 76 = _____

7. 148 + 231 = _____

8. 542 + 328 = _____

9. 475 + 267 = _____

10. 541 + 789 = _____

11. 647 – 325 = _____

12. 978 + 96 = _____

13. 1,549 – 256 = _____

Fluency Talk

How do you determine which strategy you want to use to add?

Why might you need to regroup when you subtract using an algorithm?

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